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REVIEW ARTICLE

The Role of Traditional Chinese Herbal Medicines and Bioactive Ingredients on Ion Channels: A Brief Review and Prospect

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Abstract: Traditional Chinese medicines (TCMs), particularly the Chinese herbal medicines, are valuable sources of medicines and have been used for centuries. The term “TCMs” both represents to the single drug agent like *Salvia miltiorrhiza*, *Ligusticum chuanxiong* and *Angelica sinensis*, and those herbal formulas like Jingshu Keli, Wenxin Keli and Danzhen powder. In recent years, the researches of TCMs developed rapidly to understand the scientific basis of these herbs. In this review, we collect the studies of TCM and their containing bioactive compounds, and attempt to provide an overview for their regulatory effects on different ion channels including Ca²⁺, K⁺, Na⁺, Cl⁻ channels and TRP, P2X receptors. The following conditions are used to limit the range of our review. (i) Only the herbal materials are included in this review and the animal- and mineral-original TCMs are excluded. (ii) The major discussions in this review focus on single TCM agent and the herbal formulas are only discussed for a little. (iii) Those most famous herbal medicines like *Capsicum annuum* (pepper), *Curcuma longa* (ginger) and *Cannabis sativa* (marijuana) are excluded. (iv) Only those TCM herbs with more than 5 research papers confirming their effects on ion channels are discussed in this review. Our review discusses recently available scientific evidences for TCMs and related bioactive compounds that have been reported with the modulatory effects on different ion channels, and thus provides a new ethnopharmacological approach to understand the usage of TCMs.

Keywords: Traditional Chinese medicine, Potassium channel, Calcium channel, Ion channels.

1. INTRODUCTION

Traditional Chinese medicine (TCM) represents a large group of ethnomedicines, including acupuncture, Chinese massage therapy and those pharmacotherapies with natural sources. Restricted TCM only refers to the last field, which means the decoction, pills, powders, tablets, plasters and pellets manufactured from plant, animal or mineral crude materials. The scientific study of this restricted TCM is also named “ethnopharmacology” and has received much attention in recent years.

The research interests in ethnopharmacology of TCM separately belong to two domains: one is to discover the

“single chemical entity (SCE)”, or named “bioactive compounds” derived from single medicinal plant or other natural sources, whereas the other only concentrates on establishing new therapies from the traditional usage of the natural materials and doesn’t care about the chemical basis of these natural drug mixtures. In the second research domain, the drug mixture is also called a “formula”, which represents a group of specific natural materials, with a fixed weight, proportion, and confirmed processing protocols. Some famous herbal formulas have been intensively studied, although their chemical and biochemical properties are still not yet fully understood, for example, Gosha-jinki-gan [1-3], Gua Lou Gui Zhi decoction [4, 5], Wenxin Keli [6, 7], Ba-Wei-Die-Huang-Wan [8, 9] and Yokukansan [10-13]. Due to the complex composition and confusing mechanism of herbal formula, in this short review, we only concentrate on the single medicinal plant, don’t focus on these formulas, and attempted to review the SCEs within several important herbal medicines.

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The study of natural sources to discover medicinal drugs is named “pharmacognosy”. Dried and unprepared materials, not only plants but also animals and minerals, are called “crude drugs” in all. Some animal-original crude drugs have been widely studied and their containing bioactive compounds are also divided. For instance, resibufogenin and bufalin (also named “bufanolide”) are two major ingredients in *Bufonis Venenum* derived from *Bufo bufo gargarizans* or *Bufo melanostictus*. The first one is the activator of voltage-gated Na^+ channels (VGSCs), increasing the peak amplitude and overshoot level of action potentials [14], whereas the second one is an inhibitor of L-type Ca^{2+} channel, reducing the contractility in rat ventricular myocytes [15].

It has been reported that several different peptides derived from scorpion venom act as the modulators for Na^+ channels and K^+ channels, such as iberitoxin (from scorpion *Buthus tamulus*) as the inhibitor for large-conductance calcium-activated K^+ (BK) channel and BmK toxins as the modulators for VGSCs. Here, the BmKs represent a family of neurotoxins extracted from the venom of the Asian scorpion *Buthus martensii* Karsch (BmK), including BmK I, BmK IT2, BmK AS, BmK NT1, BmK NT2, etc. Liu *et al.* reviewed different BmK toxins and their binding sites, modulatory effects on VGSCs [16]. Till now, new BmK toxins are still continuously discovered. After the publication of the above review, BmK NT1, a new α -scorpion toxin, has been reported [17, 18].

Bovis Calculus (*Bos taurus domesticus* Calculus) is another example, with cholic acid as its major ingredient. However, nowadays, most of this crude drug is replaced by the artificial type, with the same chemical components but not obtained from cattle. In this review, we only focus on those plant-original crude drugs, the animal- and mineral-original materials are only briefly discussed.

Actually, a lot of drugs or tool medicines in research are derived from plants. Some common ones like capsaicin (from *Capsicum annuum*), curcumin (from *Curcuma longa*), strychnine (from *Strychnos nux-vomica*, an antagonist of glycine and acetylcholine receptors), scopolamine (from the genus *Hyoscyamus*, *Scopolia* or *Anisodus*, a nonspecific muscarinic receptor antagonist) and cannabinoids (from *Cannabis sativa*) have been used for many years. Till now, hundreds of studies have published to describe the effects and underlying mechanisms of these drugs. Therefore, in this review, we avoid discussing these famous medicines and try to introduce the other TCMs.

There are roughly over 500 drugs listed in TCM and most of them (~85%) are plant elements [19]. To date, more than 80 TCM drugs have been reported for their regulatory effects on ion channels, and most of them (~60%) only include 1~2 papers to testify their effects. Therefore, we only select those widely reported drugs in our review, with more than 5 research works confirming their effects on ion channels. Since the 1980s, researches of TCMs, especially herbal medicines, have got vast progress. In this review, most of the crude drugs and compounds discussed have been subjected to *in vivo* studies on animals, while *in vitro* researches on cultured cells are also discussed.

2. SOME IMPORTANT TCMs

2.1. Scutellaria Baicalensis

Scutellaria baicalensis Radix (Huangqin) is the dried roots of *Scutellaria baicalensis* Georgi (Lamiaceae/Labiatae), containing several different flavonoids as major bioactive constituents, including baicalin (most important), baicalein, oroxylin A, oroxyloside, wogonin, wogonoside, norwogonoside, scutellarein and scutellarin. These flavonoids possess multiple regulatory effects on ion channels and ionotropic receptors. Baicalein, oroxylin A and wogonin were first identified as the allosteric modulators of GABA_A receptors on the benzodiazepine site [20-22]. Baicalein and wogonin are positive modulators, whereas oroxylin A is the negative type. Further investigation in HEK293T cells revealed that baicalein and baicalin selectively targeted on $\alpha 2$ and $\alpha 3$ subunit-containing GABA_A receptors [23]. These flavonoids in *Scutellaria baicalensis* can increase or decrease the Cl^- influx through the modulation on GABA_A receptors, and therefore exert some anxiolytic [24, 25] and anticonvulsant treatment effects [26]. Moreover, in these years, several reports have also suggested that oroxylin A and baicalein could promote memory and ameliorate the symptoms of schizophrenia and Alzheimer's disease through their regulation on GABA_A receptors. Liu *et al.* found that DMPC, a derivative of oroxylin A, inhibited the Cl^- influx and thus prevented against the memory impairment in a pharmacological model of schizophrenia [27], whereas Zhang *et al.* reported that baicalein reduced the production of β -amyloid ($\text{A}\beta$) by activating GABA_A receptors and increasing amyloid precursor protein (APP) α -processing [28].

Not only GABA_A receptors, the flavonoids in *Scutellaria baicalensis* also affect the function of NMDA receptors. In 2003, it is firstly reported that the aqueous extract of *Scutellaria baicalensis* blocked NMDA induced currents by 83% at a holding potential of -80 mV [29]. Further study identified four flavones including baicalin, oroxyloside, wogonoside and norwogonoside, which could bind on the second postsynaptic density 95 (PSD-95)/disc large/zonula occludens-1 (PDZ) domain to inhibit NMDA receptors [30]. Moreover, baicalin and baicalein can also block voltage-dependent Ca^{2+} channels [31] and presynaptic $\text{Ca}_v2.2$ (N-type) and $\text{Ca}_v2.1$ (P/Q-type) channels [32].

Several studies also reported the modulatory effects of flavonoids on other kinds of ion channels, including epithelial Na^+ channel (ENaC), K^+ channels and transient receptor potential channels (TRP channel family). Shih *et al.* reported that the activation of voltage-gated K^+ (K_V) channels might occur in oroxylin A-mediated relaxation of the rat smooth muscle, which was antagonized by TEA (non-selective K^+ channel blocker) and 4-AP (K_V channel blocker) [33]. Lin *et al.* then extended this conclusion and found that another flavonoid, baicalin could enhance BK channel activity in a concentration-dependent manner from 0.3~30 μM [31]. Besides, Sui *et al.* reported that baicalin down-regulated the expression of TRPV1 mRNA [34] and Deng *et al.* found that baicalin up-regulated the expression of α -ENaC protein, activated the cAMP/PKA signaling pathway and finally promoted the clearance of alveolar fluid during lipopolysaccharide (LPS)-induced immune response [35]. A previous re-

view summarized the pharmacological effects and pharmacokinetic properties of *Scutellaria baicalensis* and the containing flavones [36].

2.2. *Salvia miltiorrhiza*

Salvia miltiorrhiza Radix et Rhizoma (Danshen) is the dried rhizome and roots of *Salvia miltiorrhiza* Bunge (Lamiaceae) and it is one of the most famous TCMs for that the Compound Danshen Dripping Pills has filed the investigational new drug application to the FDA with a phase III clinical development [37]. The major usage of danshen in Chinese medicine includes treatments for cerebrovascular, cardiovascular and bleeding disorders. In 2003, Sun *et al.* firstly reported that the aqueous extract of *Salvia miltiorrhiza* reduced Ca^{2+} influx and blocked the NMDA induced currents by $52.97 \pm 1.78\%$ at a holding potential of -80 mV [29]. The same researchers repeated their results in cerebrocortical *in vitro* neurons after two years [38]. Salvianolic acid B, also named “lithospermate B”, was then identified as the first compound in *Salvia miltiorrhiza* modulating L-type Ca^{2+} channel as an antagonist [39], and this finding was supported by two parallel studies [40, 41]. Besides, salvianic acid A (also called “danshensu”) [42] and tanshinone IIA (also called “tanshinone B”) [43] were also reported as the direct blocker for voltage-gated Ca^{2+} channels (VGCC) and NMDA receptors, respectively, which reduced the Ca^{2+} influx and promoted the recovery after ischemia/reperfusion injury in middle cerebral artery occlusion (MCAO) model rats.

Tanshinone IIA is the major lipid-soluble active ingredient in *Salvia miltiorrhiza* and the major component (57.85%) among all of the tanshinones [44, 45]. Not only NMDA receptor, tanshinone IIA can also modulate different K^+ channels as an agonist. Yang *et al.* firstly reported that tanshinone IIA promoted spontaneous transient outward K^+ currents (STOCs) through the activation of BK channel [46], and a further investigation revealed that tanshinone IIA can also increase inwardly rectifying K^+ currents by activating Kir2.1 channel [47]. Ng *et al.* made a delicate *ex vivo* study and found that the blockade of K_V (by 4-AP), Kir (by Ba^{2+}) or K_{ATP} channels (by glibenclamide) exerted significant inhibition on the effects of the extract from *Salvia miltiorrhiza* and *Pueraria lobata* mixture [48]. Although the bioactive compound was not confirmed in this study, the results still clearly suggested that one or some of the ingredients in *Salvia miltiorrhiza* possessed regulatory effects on K^+ channels, including K_V , Kir and K_{ATP} types, and the antagonist of these K^+ channels also blocked the treatment effect of these components within *Salvia miltiorrhiza*.

Several other studies, only a few and contradictory, reported that salvianolic acid B and tanshinone IIA affected the function of Na^+ channel, Cl^- channel, GABA_A receptor (also permeable for Cl^- ions) and TRP channels. Yoon *et al.* investigated the effect of salvianolic acid B on Na^+ currents and action potentials in isolated rat ventricular myocytes, concluded as a novel Na^+ channel agonist [49]. Liu *et al.* [50] and Zhu *et al.* [51] reported that tanshinone IIA, the major component in essential oil, acted as the enhancer of GABA_A receptor, increasing intracellular Cl^- concentration, and the inhibitor of intracellular Cl^- channel 1 (CLIC1), decreasing

the expression of this channel, respectively. However, these contradictory consequences can not elicit any insights for that how tanshinone IIA exactly regulates Cl^- channels. Lastly, tanshinone IIA down-regulated the expression level of TRPV1 and thus exerted significant anti-nociceptive effects [52], whereas salvianolic acid B could inhibit TRPC3 and TRPC6 cation channels [53].

2.3. *Paeonia lactiflora*/*Paeonia veitchii*

Paeonia lactiflora Radix Alba (Baishao) is the dried roots of *Paeonia lactiflora* Pall. (Paeoniaceae) collected in summer and autumn, with white or creamy-white in the surface, whereas the *Paeonia veitchii* Radix Rubra (Chishao) is the dried roots of *Paeonia veitchii* Lynch (Paeoniaceae) or *Paeonia lactiflora* Pall. (Paeoniaceae) collected in spring or later autumn, with red or brown in the surface. Commonly, these two herbs are collectively referred to as “Shaoyao”, “peony” or “paeony”. Paeoniflorin is the major constituent of these herbal medicines.

In 1980s, Japanese researchers firstly identified paeoniflorin as the antagonist of nicotinic acetylcholine (nACh) receptors targeted on neuromuscular junctions [54-56]. Further investigations revealed that paeoniflorin possessed inhibitory effects on both voltage-dependent Ca^{2+} channels and ionotropic receptor-related ligand-gated Ca^{2+} channels, including L-type Ca^{2+} channels, nACh receptors and NMDA receptors. Paeoniflorin can reduce the peak amplitude of L-type Ca^{2+} current by 40.3% at the testing potential of $+10$ mV [57] and the IC_{50} value of paeoniflorin on L-type Ca^{2+} current is $14 \mu\text{M}$ [58]. Meanwhile, paeoniflorin can also inhibit NMDA receptors through binding on NR1, NR2A and NR2B subunits, specifically the NR2B type [59], as well as down-regulate the expression of NR1 mRNA [60].

Na^+ channels are the other important targets for the modulation of paeoniflorin. At the beginning, Zhang *et al.* [61] and Wang *et al.* [57] reported that paeoniflorin inhibited the Na^+ currents in acutely isolated hippocampal neurons and HEK293 cells, respectively, however, without the identification for the types of Na^+ channels. Cao *et al.* [62] and Gu *et al.* [63] then confirmed that paeoniflorin acted as the blocker of acid-sensing ion channels (ASICs, as Na^+ channels), specifically the ASIC1a channels. Besides, extracts of *Paeonia lactiflora* can also activate K_{Ca} and K_{ATP} channels to relax vascular smooth muscle [64] and down-regulate the mRNA and protein expression of TRPV1 channel to exert a significant analgesic effect on arthritis rats [65].

2.4. *Ligusticum chuanxiong*/*Angelica sinensis*

Ligusticum chuanxiong Rhizoma (Chuanxiong) is the dried rhizome of *Ligusticum striatum* var. *chuanxiong* DC. (Apiaceae) and *Angelica sinensis* Radix (Danggui) is the dried roots of *Angelica sinensis* (Oliv.) Diels (Apiaceae). Although these are two different herbs and belong to the different genus, the bioactive ingredients of them are similar. *Ligusticum chuanxiong* and *Angelica sinensis* both contain ferulic acid and ligustilide, which present as plant sterol esters. Tetramethylpyrazine, also named as ligustrazine, is the third bioactive compound and specifically exists in *Ligusticum chuanxiong*, which was reported earliest among all these three compounds.

In 2002, Tsai *et al.* firstly reported that the tetramethylpyrazine-induced vascular relaxation was inhibited by specific K^+ channel blockers that targeted on small conductance calcium-activated K^+ (SK) channels or ATP-sensitive K^+ (K_{ATP}) channels [66]. This consequence was then repeated in two other studies [67], which indicated that the vasodilation induced by tetramethylpyrazine was related to the opening of SK channels and K_{ATP} channels. Besides, tetramethylpyrazine can also block P2X3 purinoceptor related ATP-activated ion channels [68, 69], inhibit ASIC Na^+ channels [70], and suppress the Ca^{2+} influx by voltage-gated Ca^{2+} channels (VGCCs) [42].

Similarly, ferulic acid (ferulate) also possesses the ability to inhibit P2X3 receptor [69] and VGCCs [71]. Therefore, when Gu *et al.* reported that the aqueous extract of *Ligusticum chuanxiong* down-regulated the mRNA level of NMDA receptor subunit 1 (NR1), and thus might reduce the Ca^{2+} influx mediated by NMDA receptors [60], it is hard to say whether tetramethylpyrazine or ferulic acid exerted this effect, or both of them participated in the regulation of Ca^{2+} signaling.

Lastly, it comes to ligustilide. In 2011, an important study compared the modulatory effects of ligustilide (Lig) and dehydroligustilide (DH-Lig) on TRPA1 cation channel and discussed their differences [72]. According to the authors' experimental results, Lig acted as an activator and DH-Lig acted as an inhibitor to TRPA1, a mechanical and chemical sensor for pain perception [72]. However, as a common derivative of Lig, DH-Lig always exists accompanied with Lig in aqueous extract. Therefore, Lig becomes the most unclear compound among all three bioactive ingredients of *Ligusticum chuanxiong*/*Angelica sinensis*. When these two herbs or their extracts exert the treatment effect on a specific disorder, because of the opposite modulation mediated by Lig and DH-Lig, it is necessary to investigate the interaction of Lig/DH-Lig underlying this treatment process. Also, more studies are needed to reveal the effects of ligustilide on other types of ion channels.

2.5. *Pueraria lobata*

Pueraria lobata Radix (Gegen) is the dried roots of *Pueraria montana* var. *lobata* (Willd.) Ohwi (Phaseoleae). This is a herb used to treat hypertension [73] and thrombosis [74], and thus benefit to those CNS diseases like stroke, ischemia/hypoxia and cerebral hemorrhage. *Pueraria lobata* mainly consists of isoflavonoids including puerarin, daidzein and daidzin. Different to the above herbs, the ion channel targets of *Pueraria lobata* are limited to two types: K^+ channels and P2X purinoceptors (ATP-gated cation channels).

It is still debating now for the regulatory effect of puerarin on K^+ channels. At the beginning, Miao *et al.* firstly reported that puerarin inhibited the open-close rate of K^+ channel at an IC_{50} concentration between 40 μM and 80 μM [75]. However afterwards, the following studies obtained opposite conclusions, which suggested that puerarin ameliorated local blood supply through the activation of K^+ channels. Sun *et al.* found that puerarin activated BK channels and thus mediated the vasodilation [73]. Ng *et al.* then reported that the pretreatment of 4-AP (K_V channel inhibitor), Ba^{2+} (Kir channel inhibitor) and glibenclamide (K_{ATP} chan-

nel inhibitor), however not iberiotoxin (BK channel inhibitor) significantly inhibited the vasodilation induced by aqueous extract of *Pueraria lobata* [48]. These studies seem to be contradictory between each other. In 2012, Deng *et al.* concluded the previous works and found that all of the puerarin, daidzein and daidzin could exert vasorelaxant actions by opening K^+ channels [76].

After that, the tendency changed again. The study of Lee *et al.* suggested that K_{ATP} channels didn't participate in the regulatory effects of Ge-Gen-Tang (the aqueous extract of *Pueraria lobata*) on interstitial Cajal cells [77]. Similarly, the research work of Xu *et al.* also revealed that puerarin inhibited (not activated) Kir2.1, Kir2.3 and $K_{V7.1}$ channels in transfected HEK293 cells [78]. Therefore till now, it is still not clear about how puerarin, as well as daidzein and daidzin, affect the function of different K^+ channels. It may hypothesize that puerarin has different modulations on different K^+ channel types, or in different cell types. If this theory is possible, more delicate studies and careful experimental design may be needed to detect the effects of puerarin in many different conditions.

Fortunately, those studies of puerarin and P2X receptors have similar results. Puerarin performs as an antagonist for P2X3 receptors, decreasing ATP-activated currents, down-regulating the expression of P2X3 mRNA and protein, and therefore inhibiting the transmission of pain perceptions like cardiac nociceptive response [69, 79-81]. Through the above mechanisms, puerarin can reduce systolic blood pressure, decrease heart rate and relieve chest pain. These effects of puerarin make the *Pueraria lobata* as one of the fundamental Chinese herbs widely used for headaches and myocardial ischemic disorders.

2.6. *Aconitum carmichaelii*

Aconitum carmichaelii Radix, Cocta and Praeparata is the dried, cooked and salted roots of *Aconitum carmichaelii* Debeaux (Ranunculaceae), named "Chuanwu", "Zhichuanwu" and "Fuzi" in Chinese, respectively. The alkaloid toxin aconitine is the major bioactive constituent of these herbal medicines and it is always employed as an analgesic in TCM. In 1980s, aconitine has already been reported as a positive modulator for those Na^+ channels located on skeletal muscle fibers [82], neuroblastoma cells [83] and mouse brain synaptosomes [84]. However until the late 1990s, Friesen *et al.* and Gutser *et al.* identified that aconitine bound on the α subunit of VGSC protein, thus suppressed the inactivation of Na^+ channels, resulted in the Na^+ channel staying open longer, and finally prevented the repolarization of membrane potential [85, 86]. Aconitine induced an increase in synaptosomal Na^+ concentration at the EC_{50} of 3 μM and exerted an antinociceptive effect at the ED_{50} of 25-60 mg/kg [85, 86].

Interestingly, although aconitine acts as a positive modulator of voltage-dependent Na^+ channels, it can inhibit action potentials and population spikes through a mechanism of depolarizing prepulse (DPP). Due to the strong and sustained depolarization induced by aconitine, these DPPs can impair transmembrane voltage response and therefore increase the pain threshold [87]. Ameri *et al.* reported that 6-benzoylhetatisine and hetatisine, two alkaloid toxins derived from aconitine, inhibited the orthodromic and antidro-

mic population spikes in rat hippocampal slices to suppress epileptiform burst discharges [88], and Luo *et al.* also found that Shenfu decoction, an aqueous extract from the mixture of *Panax ginseng* and *Aconitum carmichaelii*, decreased the amplitude and maximum velocity of transmembrane action potentials in guinea pig ventricular myocytes to elicit anti-arrhythmic and myocardial protective effects [89].

Only a few studies reported the TRP receptor and K^+ channel mechanisms of aconitine and its derivatives apart from Na^+ channels. Wang *et al.* found that Wu-tou decoction, the aqueous extract of *Aconitum carmichaelii*, could inhibit the activities of TRPV1, TRPA1 and TRPM8 cation channels, as well as down-regulating the expression of TRPV1 and TRPA1 to achieve antinociceptive effects [90, 91], whereas Xie *et al.* also reported that hypaconitine, another alkaloid similar to aconitine, inhibited KCNH2 (hERG, $K_{v11.1}$) channels to prolong QT interval in conscious dogs [92].

2.7. *Panax ginseng*/*Panax notoginseng*

Panax ginseng Radix et Rhizoma (Renshen) is the dried rhizome and roots of *Panax ginseng* C. A. Mey. (Araliaceae) and similarly the *Panax notoginseng* Radix et Rhizoma (Sanqi) is the dried rhizome and roots of *Panax notoginseng* (Burkill) F. H. Chen (Araliaceae). They are widely used in Chinese herbal medicine and the name “*Panax*” is from the meaning “all-healing” in Greek. The triterpene saponins are the major constituents in *Panax ginseng* and *Panax notoginseng*, designated as ginsenosides (Rb1, Rd, Rg1) and notoginsenoside (R1) [93]. In 2008, the extracts from *Panax ginseng* [94] and Shenfu decoction, extracted from the mixture of *Panax ginseng* and *Aconitum carmichaelii* [89], were firstly reported as the inhibitor for VGSCs, particular those tetrodotoxin (TTX)-sensitive types. However, no further investigation supported or repeated these consequences. To other Na^+ channels, Huang *et al.* found that decichine, a non-protein diaminopropionic acid presenting in the roots of *Panax notoginseng*, exerted the coagulatory effect and shortened the tail bleeding time of rats, whereas the AMPA receptor antagonist CNQX could block this hemostatic effect [95]. Besides, Nabavi *et al.* reported that ginsenoside Rd suppressed acid-sensing ion channel (ASIC) 1a and its mediated Na^+ currents, and therefore prolonged neural cell survival during the attacks of ischemic stroke [96]. These are the current relationship between Na^+ channels and two *Panax* herbs.

Several studies suggested that the saponins from *Panax ginseng* and *Panax notoginseng* could also modulate K^+ , Ca^{2+} and Cl^- channels. Ginsenoside Rb1 could increase voltage-dependent outward K^+ current (I_{K_v}) by 27% and calcium-activated transient outward K^+ current ($I_{K_{Ca}}$) by 35%, and thus relax mouse smooth muscle [97], whereas notoginsenoside R1 contrarily inhibited voltage-gated K^+ currents in acute hippocampal slices [98]. Similarly, ginsenoside Rg1 could enhance the activity of NMDA receptors and restore long-term potentiation (LTP) impaired by morphine [99], whereas ginsenoside Rb1 contrarily down-regulated the expression of NMDA receptors and thus inhibited the influx and overload of Ca^{2+} ions [100]. Lastly, Guo *et al.* reported that ginsenoside Rb1 acted as a novel activator for the TMEM16A calcium-activated Cl^- channels (CaCCs)

at an EC_{50} of $38.4 \pm 2.14 \mu M$ [101]. Overall, different saponins may possess diverse regulations on different ion channels. To date, it is still difficult to classify those saponins with their effects clearly. Only that ginsenoside Rb1 seems to perform as an inhibitory regulator, increasing K^+ currents, down-regulating NMDA receptor expression and activating CaCCs.

2.8. *Uncaria rhynchophylla*

Uncaria rhynchophylla Ramulus cum Uncis (Gouteng) is the dried branches with uncinus of *Uncaria rhynchophylla* (Miq.) Jacks. (Rubiaceae), containing rhynchophylline (Rhy) and isorhynchophylline (Iso-rhy) as the bioactive components. These two alkaloids only target on Ca^{2+} and K^+ channels, and no other ion channels were reported. Rhy has been justified as a non-competitive NMDA antagonist ($IC_{50} = 43.2 \mu M$) and also a Ca^{2+} channel blocker [29, 102, 103]. Gui *et al.* then reported that Rhy also inhibited $K_{v11.1}$ (human ether-a-go-go-related gene, hERG) channels at an IC_{50} of $773.4 \pm 42.5 \mu M$ [104], and Zhou *et al.* reviewed those studies of Rhy on Ca^{2+} and K^+ channels until 2010 [105].

After that, several studies supported the antagonistic effects of Rhy and Iso-rhy on Ca^{2+} channels, as well as their treatment effects on heart diseases like cardiac arrhythmia and brain disorders like methamphetamine dependence. In the cardiovascular system, Iso-rhy protected against heart arrhythmias both in rats and guinea pigs through the inhibition of Ca^{2+} currents [106]. In the central nervous system, Rhy down-regulated the mRNA level [107] and protein expression [108] of NR2B subunits, and thus reversed amphetamine- or methamphetamine-induced conditioned place preference (CPP) behaviors [109].

2.9. *Evodia rutaecarpa*

Evodia rutaecarpa Fructus (Wuzhuyu) is the dried immature fruits of *Tetradium ruticarpum* (A. Juss.) T. G. Hartley (Rutaceae), former name as *Euodia ruticarpa* (A. Juss.) Benth. or *Euodia rutaecarpa* (Juss.) Benth., collected in August to November before ripeness. Indole alkaloids like evodiamine and rutaecarpine (or named rutecarpine) are the major biochemical constituents of this traditional Chinese herb. Evodiamine is suspected as the activator of TRPV1 similar to capsaicin [110] and rutaecarpine is seen as the inhibitor of cyclooxygenase 2 (COX-2) [111, 112]. Several studies supported that evodiamine acted as the partial agonist of TRPV1 weaker than capsaicin and also as a thermogenic and stimulant [113-116]. The maximum response of TRPV1 with evodiamine is 3.5-fold lower than that of capsaicin. However, interestingly, the binding of evodiamine can desensitize TRPV1 and competitively inhibit the effect of capsaicin, increasing the EC_{50} of capsaicin-activated TRPV1 currents [116]. Only one study reported the effects of indole alkaloid on K^+ and Cl^- channels. Wu *et al.* found that rutaecarpine could activate apical Cl^- channels and cAMP-dependent K^+ channels that located on rat distal colon [117].

3. SOME OF THE HERBAL FORMULAS

As we stated above, herbal formulas have complex compositions and confusing mechanisms, therefore hard to be

Table 1. Common TCMs and the number of studies about their modulatory effects on different ion channels.

Herbs	Total Papers	Ca ²⁺ Channels (Including NMDAR)	K ⁺ Channels	Na ⁺ Channels (Including AMPAR, ASICs)	Cl ⁻ Channels (Including GABA _A R)	TRP Cation Channels	P2X Receptor cation Channels
<i>Scutellaria baicalensis</i>	17	4	2	1	9	1	0
<i>Salvia miltiorrhiza</i>	15	7	3	1	2	2	0
<i>Paeonia lactiflora</i> / <i>Paeonia veitchii</i>	13	7	1	4	0	1	0
<i>Ligusticum chuanxiong</i> / <i>Angelica sinensis</i>	11	3	3	1	0	2	2
<i>Pueraria lobata</i>	10	0	6	0	0	0	4
<i>Aconitum carmichaelii</i>	10	0	1	7	0	2	0
<i>Panax ginseng</i> / <i>Panax notoginseng</i>	9	2	2	4	1	0	0
<i>Uncaria rhynchophylla</i>	8	6	2	0	0	0	0
<i>Evodia rutaecarpa</i>	6	0	1	0	1	4	0

analyzed and discussed. Here we only introduce some of the herbal formulas widely used and with relatively clear molecular mechanisms. Most of the herbal formulas can modulate, at least in part, the ionotropic receptors like AMPA, NMDA and GABA_A receptors. Gua Lou Gui Zhi decoction (GLGZD), a formula for treating cerebral ischemic stroke and related cerebral injuries, can down-regulate the expression of AMPA and NMDA receptors and reduce their activities [5]. GluR1, GluR3 and GluR4 subunits were the major types down-regulated and GluR2 was contrarily up-regulated in AMPA receptors [4], whereas the NR2A and NR2B subunits were mainly down-regulated in NMDA receptors [118]. Yokukansan (YKS) has been demonstrated to improve the behavioral and psychological symptoms of agitation, anxiety, dementia, hallucination and irritability, through antagonizing NMDA receptors [12], down-regulating NR2D subunits [13], and activating benzodiazepine (GABA_A) receptors [10]. The IC₅₀ values of YYS for the glutamate and glycine binding sites of NMDA receptors were estimated to be 37.1 and 172 µg/ml, respectively [11]. Danzhen granule, also named Danzhen powder, a formula consisting of *Salvia miltiorrhiza* and *Concha margaritifera*, is also suspected as the partial antagonist of NMDA receptors in brainstem to exert anti-migraine effects.

TRP channels and P2X receptors are also important targets for herbal formulas. Gosha-jinki-gan (GJG) is a formula treating peripheral neuropathy induced by cancer medications like oxaliplatin and paclitaxel. GJG significantly suppressed the expression of TRPA1 and TRPM8 [3], TRPV4 [2], TRPV1 and P2X3 purine receptors [1]. Similarly, another formula Ba-Wei-Die-Huang-Wan can also inhibit the expression of TRPV1 and P2X3 receptors [8, 9]. Only a few studies revealed that herbal formulas regulated voltage-dependent ion channels like VGCCs and K_v channels. Wenxin Keli, a formula used as antiarrhythmic drug, inhibited L-type Ca²⁺ currents in a concentration-dependent manner [6]. Further investigation demonstrated that calaxin (also called tridemorph), a fungicide within the components of Wenxin Keli from Gansong (*Nardostachys chinensis*), dose-

dependently blocked its predicted target Ca_v1.2 channel as the underlying mechanism [7]. Besides, another formula Jingshu Keli (JSKL) is also predicted as the modulator of K⁺ channels through the saponins from its major component *Panax notoginseng*.

CONCLUSION

Till now, there have been many studies for investigating the scientific basis of TCMs, to understand the mechanisms underlying the traditional uses of these herbal medicines. Current evidences are mainly from extensive pharmacognostic studies on both plant extracts and containing bioactive compounds, to those pharmacological studies that are limited to the tests for structural affinity and functional modulation to different ion channels. Overviewing those fundamental herbs mentioned in this review, most of them possess the inhibitory (for the nervous system) and relaxant (for cardiac system) effects through the modulation of K⁺ and Ca²⁺ channels. Ca²⁺ channels, including voltage-dependent types like L-type Ca²⁺ channels and receptor-operated ones like NMDA receptors, are the most important targets for TCMs and their ingredients, with the maximum number for related studies (29 papers). K⁺ channels are the second important targets, with 21 papers published till now (Table 1). The other four groups, Na⁺, Cl⁻ channels, TRP receptor-related cation channels and P2X receptor-related ATP-activated channels, are relatively lack of attention and reports, which reveals that these channels may not be the major modulatory targets for TCMs and their bioactive compounds.

The challenges for future investigation of TCMs for their efficacy on ion channels include two aspects: the research width and the research depth. In one hand, more works are needed to investigate other TCMs, especially those less famous herbs, for their relationship to ion channels. Additional in the other hand, it is also necessary to put more attention on Na⁺, Cl⁻ and other ion channels, detecting whether TCMs definitely don't affect on them or we just neglected these channels in the past studies. In conclusion, TCMs provide a

vast resource for bioactive constituents that may be therapeutically useful in modulating different ion channels, either in the form of aqueous and lipid extracts or as single compounds. Although the effectiveness of traditional Chinese herbal therapy is often tainted by bias, further researches should still focus on robust chemical and pharmacological experiments, to promote the development of TCMs and finally becoming one section of modern medicines.

CONSENT FOR PUBLICATION

Not applicable.

CONFLICT OF INTEREST

The authors declare no conflict of interest, financial or otherwise.

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